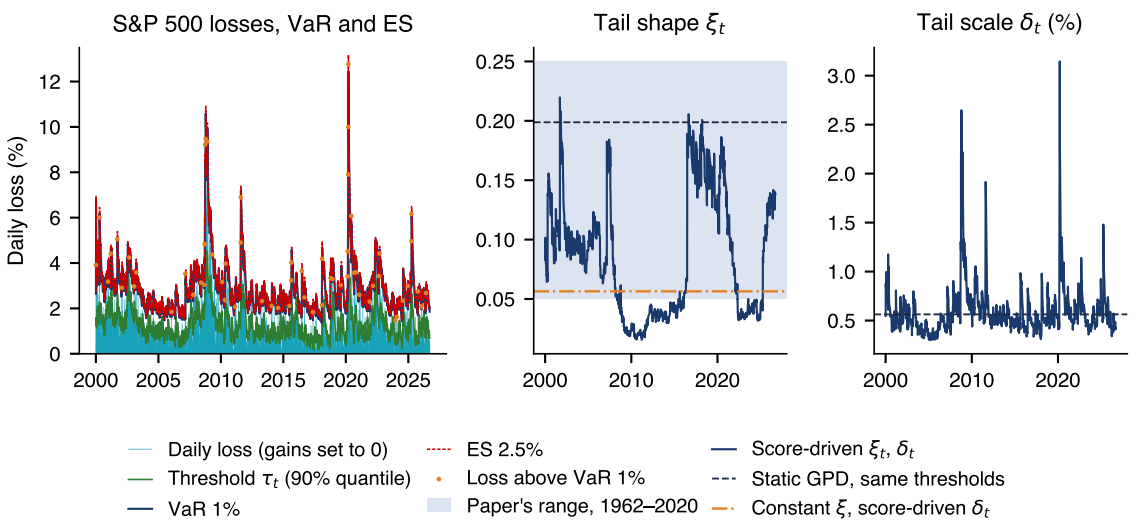

S&P 500 losses, VaR and ES
Daily loss (%)
0
2
4
6
8
10
12
2000
2005
2010
2015
2020
2025
Tail shape ξ_t
0.05
0.10
0.15
0.20
0.25
2000
2010
2020
Tail scale δ_t (%)
0.5
1.0
1.5
2.0
2.5
3.0
2000
2010
2020
Daily loss (gains set to 0)
Threshold τ_t (90% quantile)
VaR 1%
ES 2.5%
Loss above VaR 1%
Paper's range, 1962–2020
Score-driven ξ_t, δ_t
Static GPD, same thresholds
Constant ξ, score-driven δ_t